METHODS OF TEACHING

CHAPTER ONE

Introduction to Teaching and the Subject as a Discipline

Postgraduate Study Edition

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Learning Objectives

By the end of this chapter, students should be able to:

1. Define teaching and distinguish the transmission view of instruction from the facilitation (constructivist-influenced) view.
2. List and explain the basic functions a teacher performs before, during, and after instruction, and relate these to the Planning–Implementing–Assessing cycle.
3. Identify the essential characteristics of effective teachers, organized around Knowledge, Skills, and Dispositions.
4. Compare the thoughts and actions of beginning versus experienced teachers across purpose, planning, teaching, assessment, and management.
5. Explain what it means to treat a school subject as a discipline, and identify its structural elements: facts, concepts, principles, and skills.
6. Describe Pedagogical Content Knowledge (PCK) and distinguish it from content knowledge and general pedagogical knowledge.
7. Explain the TPACK framework and identify its seven constituent knowledge domains.
8. Define instructional design, identify its three core components, and compare the ADDIE, ASSURE, and Dick & Carey models.
9. Apply the structure-of-knowledge and PCK concepts to concrete examples in IT, Science, Mathematics, and Art & Design.

1. What Is Teaching?

Teaching can be defined as a deliberate, goal-directed process in which an educator organizes experiences, resources, and interactions to help learners construct knowledge, develop skills, and form attitudes they could not easily acquire on their own. Two ideas are central to this definition:

- Intentionality — teaching is planned around specific outcomes rather than left to chance.
- Facilitation — the teacher's role is to design and manage the conditions for learning, not merely to "transmit" facts from teacher to student.

1.1 Two Views of the Teacher's Role

It is useful to contrast two broad views of what a teacher does in the classroom:

View	Teacher's Role	Student's Role
Transmission view	Primary source and deliverer of knowledge	Relatively passive recipient of information
Facilitation (constructivist-influenced) view	Structures tasks, questions, and environments that guide learning	Active builder of understanding through inquiry, discussion, and problem solving

Table 1.1. Two contrasting views of the teacher's instructional role.

Most contemporary methods courses — and the InTASC (Interstate Teacher Assessment and Support Consortium) standards referenced throughout this course — favor a blend that leans toward facilitation while still recognizing that direct explanation has an important place in effective teaching.

2. Basic Teaching Functions

Regardless of subject area, effective teachers repeatedly perform a set of core functions across the instructional cycle. At the broadest level, these functions can be grouped into three overarching phases:

Phase	Description
Planning	Deciding about objectives, content, instructional strategies, student assessment, and related preparatory decisions.
Implementing	The actual delivery of the plan — presenting content, questioning, and managing the classroom in real time.
Assessing	Determining the level of student learning achieved, and using that evidence to judge progress and inform future teaching.

Table 2.1. The Planning–Implementing–Assessing macro-cycle of teaching.

At a more granular level, this macro-cycle unpacks into six recurring functions:

- Planning — setting objectives, selecting content, sequencing activities, and choosing resources before instruction begins.
- Presenting/Delivering — explaining, demonstrating, and modeling content clearly and at an appropriate pace.
- Questioning — using questions to check understanding, provoke thinking, and guide discussion.
- Managing — organizing time, materials, groupings, and behavior so that learning can proceed with minimal disruption.
- Assessing — gathering evidence of learning during and after instruction to judge progress and adjust teaching.
- Reflecting — reviewing what worked and what did not, and using that judgment to improve future lessons.

These functions are cyclical rather than strictly linear: assessment evidence from one lesson typically feeds into the planning of the next, and reflection happens continuously, not only at the end of a unit.

3. Essential Characteristics of Effective Teachers

Research on teacher effectiveness — synthesized in widely used methods texts such as Burden and Byrd's Methods for Effective Teaching (2019) — points to a recurring set of characteristics associated with strong teaching. To promote student learning and achievement, effective teachers need well-developed Knowledge, Skills, and Dispositions.

3.1 Knowledge, Skills, and Dispositions

Domain	What It Involves
1. Knowledge	The content the teacher teaches; education in general; general pedagogical knowledge; Pedagogical Content Knowledge (PCK); and Technological Pedagogical Content Knowledge (TPACK).
2. Skills	The ability to use knowledge from the areas above effectively in practice; sometimes referred to as “performance.”
3. Dispositions	Values, commitment, and professional ethics; beliefs and attitudes that are guided by, and show up in, teacher behavior.

Table 3.1. The three domains teachers draw on to promote student learning and achievement.

3.2 Research-Based Traits of Effective Teachers

Beyond this three-domain framework, effectiveness research also highlights specific observable traits:

- Solid command of subject-matter content.
- Clear, well-organized communication, both spoken and written.
- Enthusiasm and genuine interest in the subject and in learners.
- Fairness, warmth, and the ability to build rapport and trust with students.
- Strong classroom management and organizational skills.
- Flexibility — the ability to adjust plans in response to how a lesson is actually unfolding.
- A reflective, improvement-oriented professional attitude.

None of these traits operates in isolation; effective teaching is the coordinated use of several of them at once. Strong content knowledge without clear communication skills, for example, often fails to translate into student learning — which leads directly to the concept of Pedagogical Content Knowledge discussed in Section 6.

4. Beginning Teachers versus Experienced Teachers

A practical way to see how knowledge, skills, and dispositions come together in practice is to compare the typical thoughts and actions of beginning teachers with those of experienced teachers across five domains: purpose, planning, teaching, assessment, and management.

4.1 Purpose

Beginning Teachers	Experienced Teachers
• Focus on teaching subject matter. • Convey chunks of subject matter from the textbook. • Get right into details of the subject. • Cover a certain amount of material.	• Focus on teaching students. • Teach a few concepts or principles from the course syllabus. • Provide meaningful overviews of course content to be studied. • Help students grasp a few concepts well and spark interest.

Table 4.1. Contrasting sense of purpose between beginning and experienced teachers.

4.2 Planning

Beginning Teachers	Experienced Teachers
• Focus on covering textbook chapter material. • Organize lecture notes over the content. • Assume students understand what should be learned. • Plan for the students who will most likely learn what is taught.	• Identify what students may find meaningful about core concepts or principles. • Outline several instructional activities. • Specify intended learning outcomes for each lesson. • Plan to accommodate students who will learn lesson content as well as less able students.

Table 4.2. Contrasting planning practices between beginning and experienced teachers.

4.3 Teaching

Beginning Teachers	Experienced Teachers
• Attempt to transfer information to students. • Ask students to take notes and remember what they hear and see written. • Use few teaching skills and strategies.	• Engage students in thinking and finding out with questions and other means. • Ask students to explain their understandings. • Use many teaching skills and strategies.

Table 4.3. Contrasting instructional delivery between beginning and experienced teachers.

4.4 Assessment

Beginning Teachers	Experienced Teachers
• Rarely check for student learning during instruction. • Use mostly paper-and-pencil tests to determine what students know and can do. • Fail to use test results and grades to evaluate teaching effectiveness.	• Review frequently to check for student learning. • Use multiple methods to determine grades. • Consider student performance to gauge teaching effectiveness and improve instruction.

Table 4.4. Contrasting assessment practices between beginning and experienced teachers.

4.5 Management

Beginning Teachers	Experienced Teachers
• Attempt to keep students in their seats. • Make students listen and follow instructions. • Discipline students who misbehave.	• Provide opportunities for active student learning. • Give students opportunities to express their ideas. • Encourage students to monitor their own behavior.

Table 4.5. Contrasting classroom-management approaches between beginning and experienced teachers.

Taken together, Tables 4.1–4.5 illustrate a broader developmental pattern: experience tends to shift a teacher's orientation from covering content to cultivating understanding, from control to self-regulation, and from testing to using evidence formatively. This progression is not automatic — it is best supported through deliberate reflection, mentoring, and continued professional learning.

5. The Subject as a Discipline

A discipline is more than a list of topics; it has its own structure of knowledge, made up of four interrelated elements:

- Facts — specific, verifiable pieces of information.
- Concepts — categories or ideas that organize facts.
- Principles or generalizations — rules or relationships that connect concepts.
- Skills or processes — the discipline's characteristic ways of working (e.g., the scientific method in Science, algorithmic thinking in IT, proof and problem-solving in Mathematics, or the elements and principles of design in Art).

Thinking of a subject as a discipline, rather than a pile of facts to be covered, has direct consequences for instructional planning: a teacher must decide not only what content to teach but which facts, concepts, principles, and skills are most central, and in what order they should be built up so that later ideas make sense in light of earlier ones.

5.1 The Structure of Knowledge Across Four Disciplines

The table below shows how facts, concepts, principles, and skills appear differently once applied inside a specific subject.

Element	IT	Science	Mathematics	Art & Design
Fact	The keyword 'if' begins a conditional statement.	Water boils at 100°C at sea level.	7 × 8 = 56	Red, yellow, and blue are the traditional primary colors.
Concept	Variable, loop, data structure	Energy, cell, force	Function, fraction, variable	Balance, contrast, form
Principle	'Don't Repeat Yourself' (DRY) design principle	Law of conservation of energy	The commutative property of addition	Complementary colors increase visual contrast
Skill / Process	Debugging a program	Designing a controlled experiment	Solving a multi-step equation	Blending and shading technique

Table 5.1. The structure of knowledge illustrated across four teaching specializations.

6. Pedagogical Content Knowledge (PCK)

Lee Shulman (1986) introduced the influential idea of Pedagogical Content Knowledge to describe a special kind of professional knowledge unique to teachers. It is distinct from (a) Content Knowledge — knowing the subject matter itself — and (b) general Pedagogical Knowledge — knowing how to teach in general, independent of subject. PCK is the blend of the two: knowing how to represent and explain a specific idea in a specific subject so that particular learners can understand it, including awareness of common misconceptions and the most effective analogies, examples, and representations for that content.

6.1 Shulman's Knowledge Bases for Teaching

Knowledge Type	Description
Content Knowledge	Deep understanding of the subject matter itself (facts, concepts, principles).
General Pedagogical Knowledge	Broad principles of classroom management, instructional strategy, and learner development that apply across subjects.
Pedagogical Content Knowledge (PCK)	Subject-specific knowledge of how to teach a particular topic, including likely student misconceptions and effective representations.
Curricular Knowledge	Knowledge of the materials, programs, and sequencing designed to teach a given subject at a given level.

Table 6.1. Shulman's (1986) knowledge bases for teaching.

6.2 PCK in Practice Across Four Disciplines

- IT — Students commonly confuse the assignment operator (=) with the equality operator (==). Strong PCK responds by introducing assignment first through a concrete analogy — 'put this value into this labeled box' — before ever showing a conditional statement that uses ==.
- Science — Many students enter class believing heavier objects always fall faster. Strong PCK addresses this misconception directly, for example with a simulation showing a hammer and a feather falling at the same rate in a vacuum, before formally introducing Newton's second law.
- Mathematics — Many students believe multiplication always makes numbers bigger. Strong PCK deliberately chooses an early example that multiplies by a fraction less than 1, so students confront and revise this misconception before it becomes an obstacle in later algebra.
- Art & Design — Beginning students often equate 'good art' only with photographic realism. Strong PCK shows a deliberately curated range of abstract and stylized works early in the unit, broadening students' criteria for quality before asking them to critique their own work.

7. Technological Pedagogical Content Knowledge (TPACK)

TPACK describes knowledge inside educators' minds that they draw on to make instructional decisions, such as when they design lessons. Not all educators have the same TPACK, and teachers' knowledge — including TPACK — is expected to develop with more teaching experience. It extends Shulman's PCK by adding technology as a third interacting dimension alongside content and pedagogy, producing seven distinguishable, overlapping knowledge areas.

7.1 The Seven TPACK Knowledge Domains

Type of Knowledge	Definition	Example in Practice
Pedagogical Knowledge (PK)	Knowledge about the processes and practices or methods of teaching and learning.	Using scaffolding to help students' meaning-making and knowledge construction.
Content Knowledge (CK)	Knowledge of subject-matter concepts or principles.	Knowing the properties of geometric shapes.
Pedagogical Content Knowledge (PCK)	Knowledge of how to teach and represent subject matter; generating content-specific learning goals; identifying and addressing subject-specific misconceptions; content-specific assessment strategies.	Using analogical skills to teach mathematics concepts.
Technological Knowledge (TK)	Knowledge of technology hardware and software and their operation.	Using social media (e.g., blogs, wikis, Facebook).
Technological Pedagogical Knowledge (TPK)	Knowledge of how technology can support general teaching and learning activities.	Using an online quiz for assessment at the end of a lecture.
Technological Content Knowledge (TCK)	Knowledge of content-specific technologies (hardware and software) or content representations (animations or simulations).	Using virtual math manipulatives; using Google Docs to support collaborative writing.
Technological Pedagogical Content Knowledge (TPACK)	Knowledge, decision-making, and design of teaching subject matter to students with content-based technology tools and/or content-specific, technology-based assessment strategies that meet content-specific learning goals and address student misconceptions.	Using a lab where students build a ramp, collect velocity/speed data with motion detectors, then graph and interpret the relationship between velocity and speed.

Table 7.1. The seven TPACK knowledge domains, adapted from the TPACK framework.

7.2 How the Domains Converge

The three core circles — Pedagogical Knowledge (PK), Content Knowledge (CK), and Technological Knowledge (TK) — overlap pairwise to form PCK (pedagogy + content), TCK (technology + content), and TPK (technology + pedagogy). Where all three circles intersect lies TPACK itself: the simultaneous, integrated consideration of subject-matter content, instructional pedagogy, student learning, and technology, all situated within the particular classroom and school context in which the teacher works (Kelly, 2008).

More transformative technology-supported lessons tend to place the technology directly in students' hands, remain content-rich, and use technology to situate learning in ways that would be unattainable without it. The more opportunities educators have to actively design, redesign, and enact such lessons, the more their TPACK and technology-integration skills grow (Voogt et al., 2013).

8. Instructional Design

Instructional design is the practice of creating learning experiences to support learning. It is a systems approach to analyzing, designing, developing, implementing, and evaluating any instructional experience or learning product — including online courses, instructional manuals, video tutorials, and learning simulations. More specifically, instructional design is the process of using knowledge of how people learn to guide the choices of instructional sequences and strategies that meet the needs of learners and desired learning outcomes.

8.1 Three Primary Components of Instructional Design

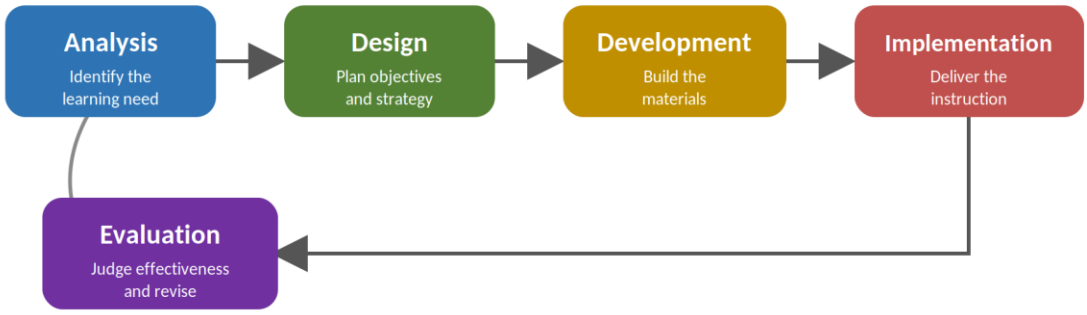
Although many models and theories guide instructional designers, instructional design practice calls for all instruction to include three primary components:

- Clear goals and objectives
- Learning activities
- Assessments

8.2 Instructional Design Models

8.2.1 ADDIE

ADDIE is one of the most well-recognized instructional design models. Founded in the 1970s at Florida State University, ADDIE represents five phases considered foundational for almost any learning design activity. The phases are typically drawn as a forward-moving sequence in which the final phase loops back into the first, making ADDIE a continuous improvement cycle rather than a one-time procedure.



Each phase feeds forward into the next, and evaluation loops back into a fresh analysis.

Figure 8.1. The ADDIE cycle: Analysis → Design → Development → Implementation → Evaluation, looping back into Analysis.

Each phase answers a distinct design question and produces a distinct output that feeds the next phase:

Phase	Key Question	Typical Activities	Output
Analysis	What is the learning problem, and who are the learners?	Needs assessment; learner-characteristics analysis; identifying goals and constraints.	A clear statement of the instructional problem and learner profile.
Design	What exactly should learners be able to do, and how will we know?	Writing learning objectives; planning assessment instruments; choosing an instructional strategy and sequence.	A design blueprint (objectives, assessments, strategy outline).
Development	What materials need to be built?	Producing learning materials, media, and activities based on the design blueprint; piloting materials on a small scale.	Draft instructional materials ready for use.
Implementation	How will this actually be delivered?	Preparing the classroom or platform; training facilitators; delivering the instruction to learners.	Instruction delivered to real learners.
Evaluation	Did it work, and how do we improve it?	Formative evaluation throughout development; summative evaluation after delivery; collecting data on learner outcomes.	Evidence-based revisions that feed back into a new Analysis phase.

Table 8.1. The five ADDIE phases in detail.

8.2.2 Applying ADDIE Across the Four Subject Specializations

Because ADDIE is a generic process model, the same five phases look different once applied inside a specific discipline. The table below illustrates one concrete application per phase for IT, Science, Mathematics, and Art & Design.

Phase	IT	Science	Mathematics	Art & Design
Analysis	Determine that students cannot yet trace how a loop updates a variable.	Determine that students hold the misconception that heavier objects fall faster.	Determine that students believe multiplication always increases a number.	Determine that students equate 'good art' only with photographic realism.
Design	Write an objective: 'trace the value of a variable through	Design a lesson objective and a formative quiz on	Sequence examples from whole-number to	Plan a critique activity comparing realistic and abstract works.

Phase	IT	Science	Mathematics	Art & Design
	three loop iterations.'	falling-object motion.	fractional multipliers.	
Development	Build a short debugging worksheet and a code-tracing table.	Prepare a vacuum hammer-and-feather simulation or video.	Develop worked examples multiplying by fractions less than 1.	Curate a gallery of abstract and stylized artworks for the unit.
Implementation	Run a live coding session with think-aloud tracing.	Show the simulation, then deliver direct instruction on Newton's second law.	Teach the sequenced examples, checking understanding after each.	Facilitate the gallery critique discussion in class.
Evaluation	Check exit-ticket tracing accuracy; revise pacing if needed.	Quiz students on the corrected misconception; revise the simulation if confusion remains.	Review quiz results; add more fraction practice if errors persist.	Assess whether students' critique criteria broadened; add more exemplars if not.

Table 8.2. Sample application of each ADDIE phase across four teaching specializations.

8.2.3 ASSURE

The ASSURE model is oriented specifically toward planning individual lessons that integrate media and technology effectively. Unlike ADDIE's linear development sequence, ASSURE is usually depicted as a cycle of six steps that a classroom teacher works through for a single lesson or unit, then revisits for the next one.



Figure 8.2. The ASSURE cycle and its six steps, arranged around the model's central purpose of guiding media-integrated lesson planning.

Each step corresponds to a practical planning decision a teacher makes when preparing a technology-supported lesson:

Step	Guiding Question	Typical Activities
A — Analyze learners	Who is the audience?	Review demographics, pedagogy/andragogy considerations, general characteristics, entry competencies, and learning styles.
S — State objectives	What do students need to learn?	Write learning-outcome assessments and course rubrics covering skill/concept, proficiency/accuracy, and conditions of performance.
S — Select methods, media, materials	What do instructors need to use?	Select instructional materials, produce new materials, or repurpose existing materials for face-to-face, hybrid, or online teaching.

Step	Guiding Question	Typical Activities
U — Utilize media, materials	How do instructors use the materials?	Preview materials, prepare the environment, and provide instruction.
R — Require learner participation	How will students actively engage?	Structure discussion, small-group activities, and formative assessment during instruction.
E — Evaluate and revise	What worked? What didn't?	Conduct summative evaluation of instructional delivery, media, and materials, and revise to improve student outcomes.

Table 8.3. The six ASSURE steps in detail.

8.2.4 Applying ASSURE Across the Four Subject Specializations

The table below shows how a teacher in each specialization might work through the same six ASSURE steps when planning one technology-supported lesson.

Step	IT	Science	Mathematics	Art & Design
Analyze learners	Note that most students already code in one language but have not used debuggers.	Note that most students hold the 'heavier falls faster' misconception.	Note that students are comfortable with whole-number multiplication only.	Note that students associate quality art with realism.
State objectives	Students will trace and correct a logic error using a debugger.	Students will predict and explain the outcome of a falling-objects experiment.	Students will compute products involving fractions less than 1.	Students will identify at least two qualities of abstract art beyond realism.
Select methods, media, materials	Choose an IDE with a visual debugger and a short buggy program.	Select a vacuum-drop simulation video and a prediction worksheet.	Select manipulatives or a virtual fraction-multiplication tool.	Select a curated slide gallery of abstract and stylized works.
Utilize media, materials	Demonstrate step-through debugging on the projector.	Play the simulation, pausing at key moments for prediction.	Model one worked example before releasing students to practice.	Display the gallery and model one sample critique aloud.
Require learner participation	Students debug a partner's code in pairs.	Students discuss predictions in small groups before revealing results.	Students solve practice problems and explain reasoning to a peer.	Students write and share a short critique of one artwork.

Step	IT	Science	Mathematics	Art & Design
Evaluate and revise	Review debugging accuracy; add more guided practice if needed.	Quiz on the corrected concept; revise pacing if misconceptions persist.	Check quiz results; reteach with additional models if needed.	Review critiques for broadened criteria; add exemplars if needed.

Table 8.4. Sample application of each ASSURE step across four teaching specializations.

9. Bringing the Concepts Together

The ideas developed in this chapter are cumulative rather than separate topics to be memorized in isolation. Teaching begins with an intentional, facilitative stance (Section 1) and is carried out through recurring functions of planning, implementing, and assessing (Section 2). Effective teachers bring together knowledge, skill, and disposition (Section 3) in ways that visibly mature with experience (Section 4). What a teacher plans for is not an arbitrary list of topics but a discipline with its own facts, concepts, principles, and skills (Section 5), and knowing how to teach that discipline to particular learners is the essence of Pedagogical Content Knowledge (Section 6). When technology becomes part of the picture, PCK expands into TPACK (Section 7). Finally, instructional design provides the systematic process — goals, activities, and assessments, organized through models such as ADDIE, ASSURE, or Dick and Carey — through which all of the above is translated into an actual lesson or course (Section 8).

For postgraduate study, it is worth noting that these frameworks are not competing theories to be chosen between, but complementary lenses: Shulman's PCK explains why subject-specific pedagogical expertise matters; TPACK extends that explanation into technology-rich environments; and instructional design models supply the concrete procedural steps for turning pedagogical judgment into a deliverable learning experience.

Chapter Summary

Teaching is an intentional, facilitative process built around a recurring cycle of planning, presenting, questioning, managing, assessing, and reflecting. Effective teachers combine strong subject knowledge with communication skills, management ability, fairness, and a reflective disposition — and these qualities visibly develop as teachers move from beginning to experienced practice.

Treating a subject as a discipline — with its own facts, concepts, principles, and skills — rather than a list of topics helps teachers plan more coherent instruction. Shulman's concept of Pedagogical Content Knowledge captures the specialized expertise teachers need at the intersection of subject-matter expertise and teaching skill; the TPACK framework extends this expertise into technology-integrated teaching contexts. Instructional design, in turn, supplies the systematic process — clear goals, learning activities, and assessments, organized through models such as ADDIE, ASSURE, or Dick and Carey — by which this pedagogical knowledge is turned into effective lessons and courses. Together, these concepts set the stage for the planning and standards topics covered in the next chapter.

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